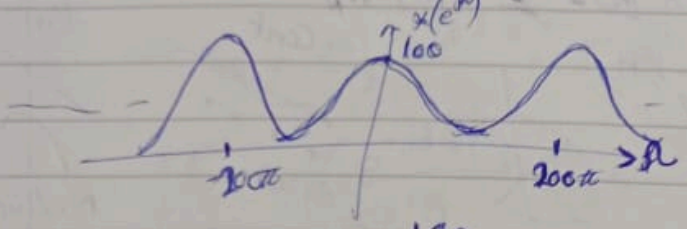
Assignment 2

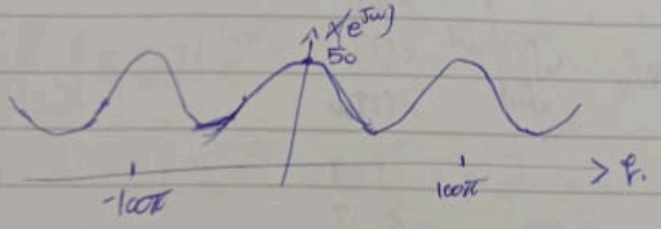
Q1.

$$X(e^{j\Omega}) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X_a(j(\Omega - k\Omega_s)) \qquad \Omega_s = 2\pi f_s.$$

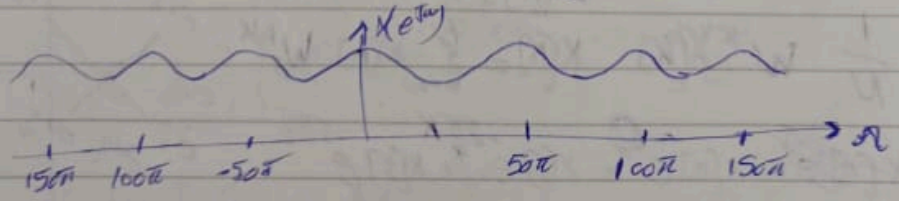
a) $X(e^{j\omega}) = F_s \sum_{k=-\infty}^{\infty} \frac{100}{100 + (\Omega - k2\pi f_s)^2} = 100 \sum \frac{100}{100 + (\Omega - 200\pi k)^2}$



b) $X(e^{j\omega}) = 50 \sum \frac{100}{100 + (\Omega - 100\pi k)^2}$



c) $X(e^{j\omega}) = 25 \sum \frac{100}{100 + (\Omega - 50\pi k)^2}$



d) $\frac{100}{100 + \Omega^2} = 0.01$ ← 1% $\Rightarrow \Omega = 99.5$ rad/s.

$F_{max} = 15.8$ Hz.

$2F_{max} = 31.6$

a) 100 Hz > $2f_{max}$ ✓

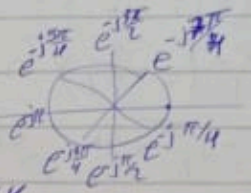
b) 50 Hz > $2f_{max}$ ✓

c) 25 Hz. ✗

Q2

a) $x[n] = 4-n, \quad N=8$

$$X[k] = \sum_{n=0}^{7} x[n] W_8^{nk}$$



$$X[k] = 4\times1 + 3\times e^{-j\pi/4 k} + 2e^{-j\pi/2 k} + e^{-j3\pi/4 k} + 0\times e^{-j\pi k} - e^{-j5\pi/4 k} - 2e^{-j6\pi/2 k} - 3e^{-j7\pi/4 k}$$

$$e^{-j\frac{5\pi}{4}k} = e^{2\pi k - j\frac{5\pi}{4}k} = e^{j3\pi/4 k}$$

So $X[k] = 4 - 6j\sin\frac{k\pi}{4} - 4j\sin\frac{k\pi}{2} - 2j\sin\frac{3k\pi}{4}$

b) $x[n] = 4\sin 0.2\pi n \quad N=10$

$$x[n] = \frac{2}{j}\left[e^{0.2\pi nj} - e^{-0.2\pi nj}\right]$$

$$X[k] = \frac{2}{j}\sum_{n=0}^{9}\left[e^{\frac{2\pi nj}{10}} - e^{j\frac{2\pi n}{10}}\right]e^{-j\frac{2\pi}{10}nk}$$

$$= \frac{2}{j}\sum_{n=0}^{9} e^{\frac{2\pi nj}{10}} e^{-j\frac{2\pi}{10}nk} - \frac{2}{j}\sum_{n=0}^{9} e^{-\frac{2\pi nj}{10}} e^{-j\frac{2\pi}{10}nk}$$

orthogonal. orthogonal.

$$\sum_{n=0}^{N-1} e^{-j\frac{2\pi}{N}nk} e^{j\frac{2\pi}{N}nm} = N\,\delta[k-m]$$

$$X[k] = \frac{2}{j}\left[10\,\delta[k-1] - 10\,\delta[k+1]\right]$$

$$= \frac{2}{j}\left[10\,\delta[k-1] - 10\,\delta[k-9]\right]$$

c) $x[n] = 6\cos^2(0.2\pi n) \quad N=10$

$$x[n] = 6\left[\frac{1}{2} + \frac{1}{2}\cos 0.4\pi n\right] = 3 + 3\cos 2\frac{2\pi}{10}n$$

$$= 3 + \frac{3}{2}e^{j\frac{2\pi}{10}2n} + \frac{3}{2}e^{-j\frac{2\pi}{10}2n}$$

$$x[n] = \frac{1}{N}\left[X[0] + X[1]e^{j\frac{2\pi}{N}k(1)} + X[2]e^{j\frac{2\pi}{N}k(2)} + \cdots X[n-1]e^{j\frac{2\pi}{N}k(n-1)}\right]$$

By comparing

$X[0] = 10 \times 3 = 30$

$x[2] = 3/2 \times 10 = 15$

$e^{-j\frac{2\pi}{10} \times 2n} = e^{2\pi n j - j\frac{2\pi}{10} \times 2n} = e^{j\frac{2\pi}{10} 8n}$

$x[8] = 3/2 \times 10 = 15$

$x[k] = 30\,\delta[k] + 15[\,\delta[k-2] + \delta[k-8]\,]$

d) $x[n] = 5(0.8)^n \quad N = 16$

$x[k] = \sum_{n=0}^{15} 5(0.8)^n e^{-j\frac{2\pi}{16}kn} = \sum_{n=0}^{15} 5\,(0.8\,e^{-j\frac{2\pi}{16}k})^n$

$=$ → 16×5, $0.8\,e^{-j\frac{2\pi}{16}k} = 1$ refused.

→ $5 \times \frac{1-0.8^{16}\,e^{-j2\pi k}}{1-0.8\,e^{-j\frac{2\pi}{16}k}} = 5\,\frac{1-0.8^{16}}{1-0.8\,e^{-j\frac{2\pi}{16}k}}$

e) $x[n] = \begin{cases} 3 & \text{even } n \\ -2 & \text{odd } n. \end{cases} \quad N = 20$

$x[k] = \sum_{n=0}^{19} x[n]\, e^{-j\frac{2\pi}{20}kn}$

even $n \rightarrow n = 2m$

odd $n \rightarrow n = 2m+1$

$x[k] = \underbrace{\sum_{n=0}^{9} 3\,e^{-j\frac{2\pi}{20}2mk}}_{x_1[k]} - 2\underbrace{\sum_{n=0}^{9} e^{-j\frac{2\pi}{20}(2m+1)k}}_{x_2[k]}$

$x_1[k] = \begin{cases} 3 \times 10\ e^{-j\frac{2\pi}{20}\times 2k} = 1 & k = 0, 10, 20, \dots \\ \frac{1-e^{-j2\pi k}}{1-e^{-j\frac{2\pi}{20}k}} = 0 & \text{, otherwise.} \end{cases}$

$x_1[k] = 30\,\delta[k]$

$$X_2[k] = \begin{cases} (-2\times 10)\, e^{-j\frac{2\pi}{20}k} \cdot \dfrac{e^{-j\frac{2\pi}{20}\times 2k}}{e^{-j\frac{2\pi}{20}k}} \cdot \dfrac{1-e^{-j2\pi k}}{1-e^{-j\frac{2\pi}{20}k}} & \text{, } k=0,10,\ldots \\ 0 & \text{, otherwise} \end{cases}$$

$e^{-j2\pi k} = 1$ (1)

$$x[k] = 30\,\delta[k] + 30\,\delta[k-10] - 20\,\delta[k] - 20\,\delta[k-10]$$

$$x[k] = 10\,\delta[k] + 50\,\delta[k-10]$$

Q3

$x_1[n] \circledast x_2[n]$ $\quad x_1[n] = \{1,2,3,4,5\}$ $\quad N_1 = 5.$

$x_2[n] = \{2,-1,1,-1\}$ $\quad N_2 = 4.$

a) $x_2[n] = \{2,-1,1,-1,0\}$

$$\begin{bmatrix} y[0] \\ y[1] \\ y[2] \\ y[3] \\ y[4] \end{bmatrix} = \begin{bmatrix} 2 & 0 & -1 & 1 & -1 \\ -1 & 2 & 0 & -1 & 1 \\ 1 & -1 & 2 & 0 & -1 \\ -1 & 1 & -1 & 2 & 0 \\ 0 & -1 & 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{bmatrix}$$

$y[0] = -2$ $\quad y[3] = 6$

$y[1] = +4$ $\quad y[4] = 7.$

$y[2] = 0$

$y[n] = \{-2, 4, 0, 6, 7\}$

$W_5 = e^{-j\frac{2\pi}{5}}$

c) $X_1[k] = \sum_{n=0}^{4} x_1[n]\, W_5^{kn} = 1 + 2W_5^{k} + 3W_5^{2k} + 5W_5^{4k}$

$X_2[k] = \sum_{n=0}^{4} x_2[n]\, W_5^{kn} = 2 - W_5^{k} + W_5^{2k} - W_5^{3k} + 0\,W_5^{4k}$

for $k=0$ $\quad X_1[0] = 15$ $\quad X_2[0] = 1.$

$Y[k] = X_1[k]\, X_2[k]$

$k=0$ $\quad Y[0] = 15$

IDFT

$$y[n] = \frac{1}{5} \sum_{k=0}^{4} Y[k] W_5^{-kn}$$

$$\sum y[n] = -2+4+0+6+7 = 15 = y[0] = 15.$$ DC

Q4

a) $y[n] = x_1[n] \circledast x_2[n]$ $\quad y[n]_{length} = 4+6-1 = 9$

$$\begin{bmatrix} y[0] \\ y[1] \\ y[2] \\ y[3] \\ y[4] \\ y[5] \\ y[6] \\ y[7] \\ y[8] \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ -3 & 1 & -2 & 1 & 0 & 0 \\ 0 & -3 & 1 & -2 & 1 & 0 \\ 0 & 0 & -3 & 1 & -2 & 1 \\ 0 & 0 & 0 & -3 & 1 & -2 \\ 0 & 0 & 0 & 0 & -3 & 1 \\ 0 & 0 & 0 & 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ -1 \\ 0 \\ 0 \\ 4 \end{bmatrix}$$

$y[0] = 0$ $\quad y[4] = -6-1 = -7$ $\quad y[8] = -12$

$y[1] = 2$ $\quad y[5] = 3+4 = 7$

$y[2] = -4-1 = -5$ $\quad y[6] = -8$

$y[3] = 2+2 = 4$ $\quad y[7] = 4$

b) $y_c[n] = \sum_{k=-\infty}^{\infty} y_L[n+6k]$ $\quad n = 0,1,2,3,4,5$

$y_c[0] = y_L[0] + y_L[6] = 0 + (-8) = -8.$

$y_c[1] = y_L[1] + y_L[7] = 2+4 = 6$

$y_c[2] = y_L[2] + y_L[8] = -5 + (-12) = -17$

$y_c[3] = y_L[3] = 4$

$y_c[4] = y_L[4] = -7$ $\quad y_c[5] = y_L[5] = 7.$

$$y_c[n] = \{-8, 6, -17, 4, -7, 7\}$$

c) $L_{linear} = 9$

$N \geq 9$

$N = 9$
Smallest value

$x_2[0] = d = x_1[5]$
$x_2[2] = c = x_1[3]$
$x_2[3] = b = x_1[2]$
$x_2[5] = a = x_1[0]$

$n \rightarrow 5-n \mod 8$

$$x_2[k] = W_8^{5k} V[k]$$

$$x_2[k] = W_8^{5k} x_1[(-k) \bmod 8]$$

$x_1[n] = \{1, 0, 2, 3, 0, 4, 0, 0\}$
$x_2[n] = \{4, 0, 3, 2, 0, 1, 0, 0\}$

$$X_1[k] = 1 + 2W_8^{2k} + 3W_8^{3k} + 4W_8^{5k}$$
$$X_2[k] = 4 + 3W_8^{2k} + 2W_8^{3k} + 1W_8^{5k}$$

$$X_1[-k] = 1 + 2W_8^{-2k} + 3W_8^{-3k} + 4W_8^{-5k}$$

$$W_8^{5k} X_1[-k] = W_8^{5k}(1 + 2W_8^{-2k} + 3W_8^{-3k} + 4W_8^{-5k})$$

$$= W_8^{5k} + 2W_8^{3k} + 3W_8^{2k} + 4W_8^{0}$$
$$= 4 + 3W_8^{2k} + 2W_8^{3k} + W_8^{5k}$$

Q6

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{nk}$$

a)

$$X[k] = \sum_{n=0}^{\frac{N}{2}-1} x[n] W_N^{nk} + \sum_{n=N/2}^{N-1} x[n] W_N^{nk} \qquad W_N = e^{-j\frac{2\pi}{N}}$$

$m = n - N/2 \qquad n = m + N/2$

$$\sum_{n=N/2}^{N-1} x[n] W_N^{nk} = \sum_{m=0}^{N/2-1} x\left[m+\tfrac{N}{2}\right] W_N^{(m+N/2)k}$$

$$W_N^{\frac{N}{2}k} = \left(e^{-j\frac{2\pi}{N}}\right)^{\frac{N}{2}k} = e^{-j\pi k} = (-1)^k$$

$$\sum_{m=0}^{\frac{N}{2}-1} -x[m] W_N^{mk} (-1)^k = -(-1)^k \sum_{m=0}^{\frac{N}{2}-1} x[m] W_N^{mk}$$

$$x[k] = \sum_{n=0}^{\frac{N}{2}-1} x[n] W_N^{nk} - (-1)^k \sum_{n=0}^{\frac{N}{2}-1} x[n] W_N^{nk}$$

$$x[k] = \left(1 - (-1)^k\right) \sum_{n=0}^{\frac{N}{2}-1} x[n] W_N^{nk}$$

even $k = 2m \quad (-1)^k = 1$

$$X[k] = (1-1) \sum_{n=0}^{\frac{N}{2}-1} x[n] W_N^{nk} = 0$$

b) $N = 4m \quad x[n] = -x\left[\langle n + N/4 \rangle_N\right]$

$$x\left[n+\tfrac{N}{4}\right] = -x[n]$$

$$x\left[n+\tfrac{2N}{4}\right] = -x\left[n+\tfrac{N}{4}\right] = -(-x[n]) = x[n]$$

$$x\left[n+\tfrac{3N}{4}\right] = -x\left[n+\tfrac{2N}{4}\right] = -x[n]$$

$$x[k] = \sum_{n=0}^{\frac{N}{4}-1} x[n] W_N^{nk} + \sum_{n=N/4}^{\frac{2N}{4}-1} x[n] W_N^{nk} + \sum_{n=\frac{2N}{4}}^{\frac{3N}{4}-1} x[n] W_N^{nk} + \sum_{n=\frac{3N}{4}}^{N-1} x[n] W_N^{nk}$$

$$= \sum_{n=0}^{\frac{N}{4}-1} x[n] W_N^{nk} + x\left[n+\frac{N}{4}\right] W_N^{\left(n+\frac{N}{4}\right)k} + x\left[n+\frac{2N}{4}\right] W_N^{\left(n+\frac{2N}{4}\right)k}$$

$$+ x\left[n+\frac{3N}{4}\right] W_N^{\left(n+\frac{3N}{4}\right)k}$$

$$W_N^{P\frac{N}{4}K} = \left(e^{-j\frac{2\pi}{4}}\right)^{P\frac{N}{4}K} = (-j)^{PK}$$

$$X[K] = \sum_{n=0}^{\frac{N}{4}-1} x[n] W_N^{nK} \underbrace{\left[1-(-j)^K+(-j)^{2K}-(-j)^{3K}\right]}_{A}$$

$K = 4l$

$$(-j)^K = (-j)^{4l} = (1)^l = 1$$

$$A(4l) = 1 - (1) + (1)^2 - (1)^3$$

$$= 0$$

$$X[4l] = 0$$

Q7 $\omega_p = 0.3\pi$, $A_P = 0.5dB$

$\omega_s = 0.5\pi$, $A_s = 50dB$

$\Delta\omega = 0.2\pi$ $\quad \omega_c = \frac{0.3\pi + 0.5\pi}{2} = 0.4\pi$

$\delta_P = 0.0287$

$\delta_s = 3.162 \times 10^{-3}$ $\rightarrow$ $\delta = \delta_s = 3.162 \times 10^{-3}$

$A = A_s = 50\,dB$.

$\frac{6.6\pi}{L} = 0.2\pi$ $\rightarrow$ $L = 33$ $\rightarrow$ $M = 32$

$\hookrightarrow$ type I

$N = 16$

$h[n] = \frac{\sin(\omega_c(n-\alpha))}{\pi(n-\alpha)} \times$ hamming window of length 33.

```
% Define the sequences
x1 = [1, 2, 3, 4, 5];
x2 = [2, -1, 1, -1];
N = 5;
% using MATLAB's built-in circular convolution:
y = cconv(x1, x2, N);

disp('Circular Convolution Result:');
disp(y);
```

```
>> ver_1
Circular Convolution Result:
   -2.0000    4.0000    0.0000    6.0000    7.0000

>>
```

```
%% FIR Filter Design Specifications
wp = 0.3 * pi;  % Passband edge frequency (rad/sample)
ws = 0.5 * pi;  % Stopband edge frequency (rad/sample)
Ap = 0.5;       % Passband ripple (dB)
As = 50;        % Minimum stopband attenuation (dB)

% Ideal cutoff frequency is the midpoint of the transition band
wc = (wp + ws) / 2;
% Transition bandwidth
delta_w = ws - wp;
```

```
%% (a) Fixed Window Design
% To achieve a stopband attenuation of at least 50 dB, we need a Hamming window.
% (Rectangular = ~21dB, Bartlett = ~25dB, Hanning = ~44dB, Hamming = ~53dB)

% The empirical transition width for a Hamming window is approx 8*pi / N.
% Therefore, the required length N is:
N_hamming = ceil(8 * pi / delta_w);

% Ensure the filter length is odd (even order) to create a Type I linear phase filter
if mod(N_hamming, 2) == 0
    N_hamming = N_hamming + 1;
end
order_hamming = N_hamming - 1;

% Design the filter using fir1
% Note: fir1 takes normalized frequency where 1.0 corresponds to pi
h_fixed = fir1(order_hamming, wc/pi, 'low', hamming(N_hamming));
```

```
%% (b) Kaiser Window Design
% MATLAB provides 'kaiserord' to automatically compute the optimal filter
% order and beta parameter based on the exact ripple specifications.

% Convert frequencies to normalized values (0 to 1, where 1 corresponds to pi)
f_edges = [wp/pi, ws/pi];
mags = [1 0]; % Desired amplitudes (1 for passband, 0 for stopband)
```

```
% Convert dB specifications to linear deviations
dev_p = (10^(Ap/20) - 1) / (10^(Ap/20) + 1);
dev_s = 10^(-As/20);
devs = [dev_p, dev_s];

% Calculate Kaiser parameters
[order_kaiser, Wn, beta, ftype] = kaiserord(f_edges, mags, devs);
N_kaiser = order_kaiser + 1;

% Design the filter
h_kaiser = fir1(order_kaiser, Wn, ftype, kaiser(N_kaiser, beta));
```

```
%% Output Comparison to Command Window
fprintf('--- Filter Length Comparison ---\n');
fprintf('Fixed Window (Hamming) Length: %d\n', N_hamming);
fprintf('Kaiser Window Length: %d\n', N_kaiser);
fprintf('--------------------------------\n');
```

```
%% Plotting the Results
% Use freqz to compute the frequency response (1024 points for smooth curves)
[H_fixed, w] = freqz(h_fixed, 1, 1024);
[H_kaiser, ~] = freqz(h_kaiser, 1, 1024);

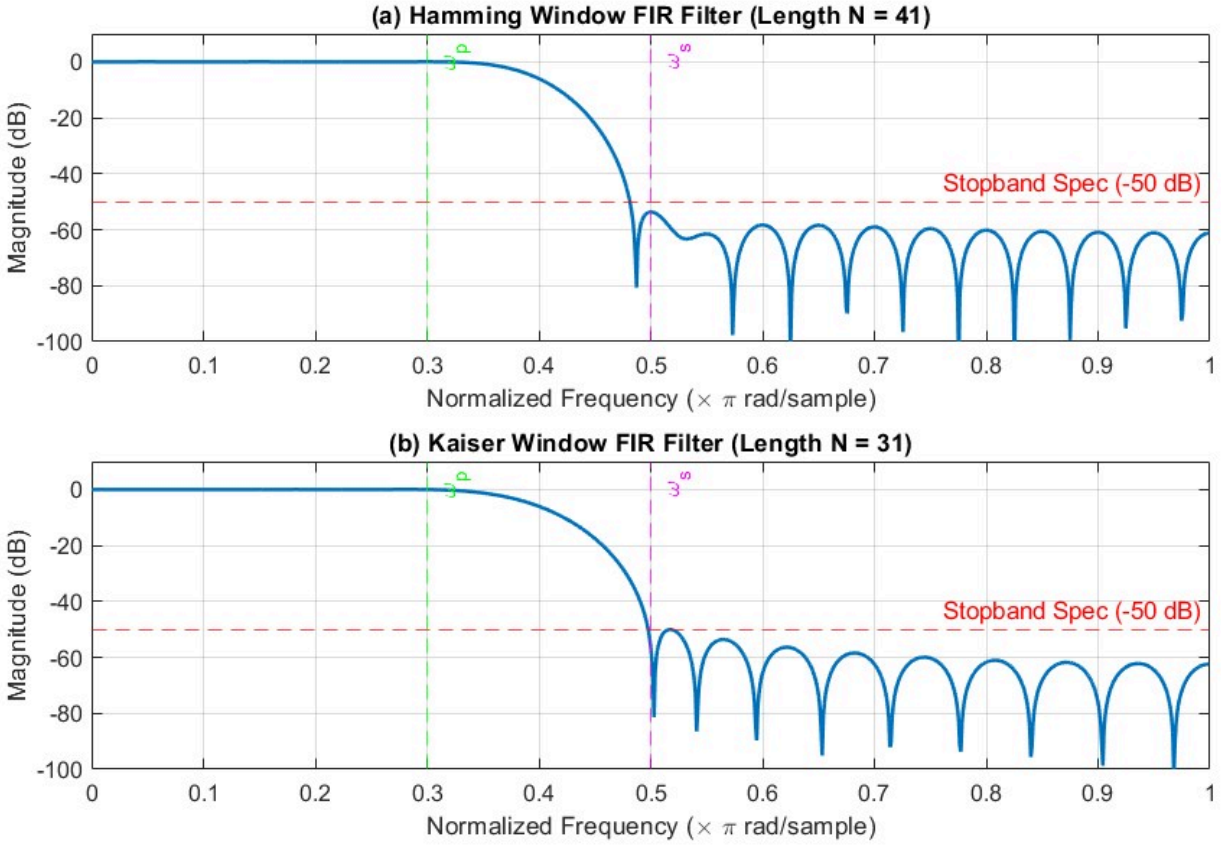
% Convert magnitude to Decibels
mag_fixed_dB = 20*log10(abs(H_fixed));
mag_kaiser_dB = 20*log10(abs(H_kaiser));

figure('Name', 'FIR Filter Design Comparison', 'Position', [100, 100, 800, 500]);

% Plot Fixed Window (Hamming)
subplot(2,1,1);
plot(w/pi, mag_fixed_dB, 'LineWidth', 1.5);
hold on;
```

```
yline(-As, 'r--', 'Stopband Spec (-50 dB)');
xline(wp/pi, 'g--', '\omega_p');
xline(ws/pi, 'm--', '\omega_s');
hold off;
title(['(a) Hamming Window FIR Filter (Length N = ', num2str(N_hamming), ')']);
ylabel('Magnitude (dB)');
xlabel('Normalized Frequency (\times \pi rad/sample)');
grid on;
ylim([-100 10]);

% Plot Kaiser Window
subplot(2,1,2);
plot(w/pi, mag_kaiser_dB, 'LineWidth', 1.5);
hold on;
yline(-As, 'r--', 'Stopband Spec (-50 dB)');
xline(wp/pi, 'g--', '\omega_p');
xline(ws/pi, 'm--', '\omega_s');
hold off;
title(['(b) Kaiser Window FIR Filter (Length N = ', num2str(N_kaiser), ')']);
ylabel('Magnitude (dB)');
xlabel('Normalized Frequency (\times \pi rad/sample)');
grid on;
ylim([-100 10]);
```


(a) Hamming Window FIR Filter (Length N = 41)
Magnitude (dB)
0
-20
-40
-60
-80
-100
ω_p
ω_s
Stopband Spec (-50 dB)
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (× π rad/sample)

(b) Kaiser Window FIR Filter (Length N = 31)
Magnitude (dB)
0
-20
-40
-60
-80
-100
ω_p
ω_s
Stopband Spec (-50 dB)
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (× π rad/sample)